242672: felsic metavolcanic rock, Solomon Road

(Youanmi Terrane, Yilgarn Craton)

Location and sampling

Perenjori (SH 50-6), Perenjori (2139)
MGA Zone 50, 437078E 6750471N

Sampled on 5 May 2019

This sample was collected from low outcrop at the northern edge of a wooded area, about 9.7 km northeast of Perenjori townsite, 3.7 km west-northwest of Ryan Bore, and 0.07 km west of Solomon Road.

Geological context

The unit sampled is a metarhyolite in the southern part of the Koolanooka greenstone belt, which consists mainly of greenschist to amphibolite facies volcanic, sedimentary, and ultramafic to felsic intrusive rocks, and banded iron-formation (Baxter et al., 1983; Baxter and Lipple, 1985; Watkins, 1990). A felsic metavolcanic rock, sampled in the northern Koolanooka greenstone belt, about 15.9 km to the north-northwest, yielded an igneous crystallization age of 2961 ± 4 Ma and an inherited zircon component dated at 2994 ± 3 Ma (GSWA 242652, Wingate et al., 2021). A metamonzogranite collected about 32 km to the north yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020).

Petrographic description

The sample is a very fine-grained, compositionally layered, felsic metavolcanic rock (Fig. 1), consisting of about 20–25% plagioclase, 20–25% K-feldspar, 25% garnet, 20% quartz, 6–7% epidote, 3–4% clinopyroxene, and minor sericite, zircon, apatite and carbonate minerals. The layering is defined mainly by garnet and lenses of quartz, and by subtle differences in grain size within quartzofeldspathic parts of the sample. Quartzofeldspathic layers are composed of mainly equant, polygonal grains between 0.05 and 0.2 mm in size. Plagioclase is identified by polysynthetic twinning and alteration to sericite and minor carbonate minerals, whereas K-feldspar exhibits microcline twinning. Some coarser-grained feldspar aggregates form elongate lenses up to 4 mm long. Quartz is equant, transparent, partially in poorly defined lenses, and also forms tiny round inclusions within K-feldspar. Colourless to pale pink garnet forms porphyroblasts in lenticular patches up to 6 mm long that have poorly defined, irregular margins and are intergrown with quartzofeldspathic layers and, to a much lesser extent, with clinopyroxene. There is a 1.5 mm-wide band of fractured garnet with minor quartz, and also several narrower lenses of fractured garnet. Garnet is partially replaced by irregular patches of epidote up to 1 mm in size. Clinopyroxene is colourless to pale green and forms mainly equant, anhedral grains up to 0.5 mm in diameter, and less common elongate grains up to 1 mm long. Most clinopyroxene is disseminated throughout the quartzofeldspathic aggregate, although some is associated with garnet. Apatite is subhedral, tabular, and up to 0.1 mm in size. The sample is strongly recrystallized, and the garnet–clinopyroxene–plagioclase–K-feldspar assemblage is consistent with high-grade metamorphism of a fine-grained felsic volcanic protolith.

Zircon morphology

Zircons isolated from this sample are pale pink to dark red and subhedral to euhedral. The crystals are up to 200 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, broad concentric zoning is ubiquitous, and some crystals exhibit narrow, discontinuous rims of luminescent material. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 16–17 October 2020, using SHRIMP-B. Nine analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.26% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.24%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 17 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Figure 1. **Hand specimen image of sample 242672: felsic metavolcanic rock, Solomon Road**

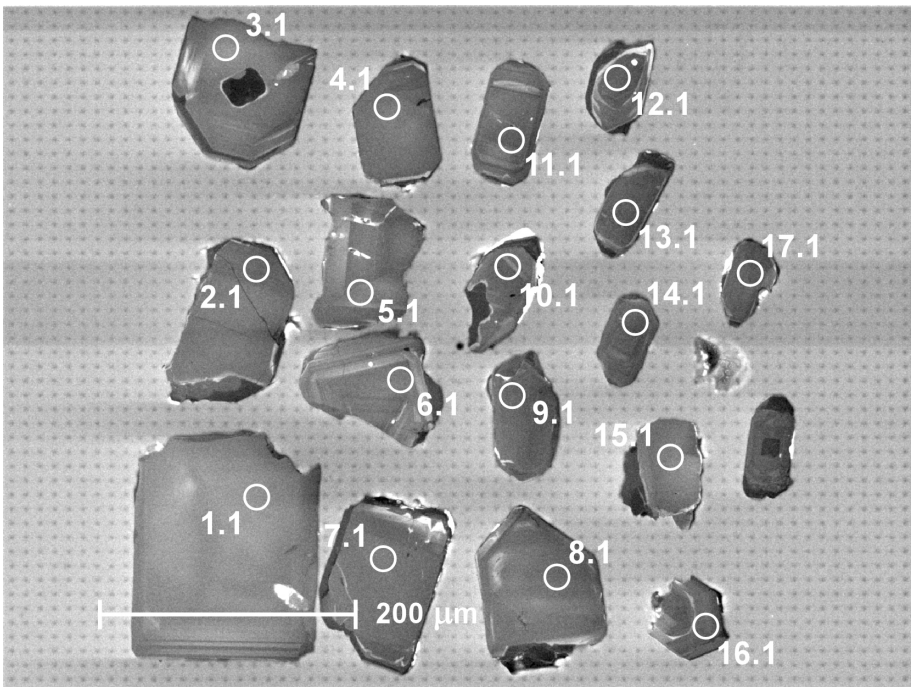

Figure 2. **Cathodoluminescence image of all zircons from sample 242672: felsic metavolcanic rock, Solomon Road. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 242672: felsic metavolcanic rock, Solomon Road

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	3	3.1	286	216	0.75	0.000	1.712	0.017	0.22096	0.00117	1.712	0.017	0.22096	0.00117	2966	24	2988	9	0.7
I	4	4.1	299	224	0.75	0.017	1.723	0.017	0.22171	0.00116	1.723	0.017	0.22156	0.00117	2951	24	2992	8	1.4
I	5	5.1	217	138	0.64	0.036	1.711	0.019	0.22190	0.00130	1.711	0.019	0.22158	0.00131	2966	26	2992	10	0.9
I	10	10.1	239	157	0.66	-0.012	1.682	0.034	0.22200	0.00128	1.682	0.034	0.22211	0.00129	3008	48	2996	9	-0.4
I	6	6.1	253	182	0.72	0.039	1.739	0.018	0.22274	0.00121	1.739	0.018	0.22239	0.00122	2928	25	2998	9	2.3
I	11	11.1	355	393	1.11	0.123	1.881	0.018	0.22372	0.00115	1.884	0.018	0.22264	0.00119	2745	22	3000	9	8.5
I	7	7.1	285	197	0.69	-0.026	1.748	0.017	0.22253	0.00117	1.747	0.017	0.22276	0.00118	2918	24	3001	9	2.8
I	9	9.1	277	167	0.60	0.050	1.727	0.018	0.22351	0.00122	1.727	0.018	0.22307	0.00124	2944	25	3003	9	2.0
I	13	13.1	353	239	0.68	0.000	1.706	0.016	0.22319	0.00111	1.706	0.016	0.22319	0.00111	2974	23	3004	8	1.0
I	16	16.1	224	151	0.67	-0.011	1.749	0.019	0.22310	0.00126	1.748	0.019	0.22319	0.00126	2916	25	3004	9	2.9
I	15	15.1	182	96	0.53	-0.029	1.755	0.020	0.22308	0.00140	1.754	0.020	0.22334	0.00141	2908	27	3005	10	3.2
I	2	2.1	323	276	0.86	-0.039	1.713	0.017	0.22313	0.00113	1.712	0.017	0.22347	0.00114	2966	24	3006	8	1.3
I	1	1.1	201	135	0.67	-0.049	1.713	0.019	0.22310	0.00131	1.712	0.019	0.22354	0.00133	2966	27	3006	10	1.4
I	14	14.1	318	196	0.61	0.016	1.708	0.017	0.22375	0.00114	1.708	0.017	0.22361	0.00114	2971	24	3007	8	1.2
I	12	12.1	362	284	0.79	0.039	1.714	0.017	0.22412	0.00113	1.714	0.017	0.22378	0.00114	2962	23	3008	8	1.5
I	8	8.1	372	294	0.79	0.028	1.705	0.016	0.22418	0.00110	1.706	0.016	0.22393	0.00111	2974	23	3009	8	1.2
I	17	17.1	340	285	0.84	-0.017	1.707	0.017	0.22426	0.00116	1.707	0.017	0.22440	0.00116	2973	24	3013	8	1.3

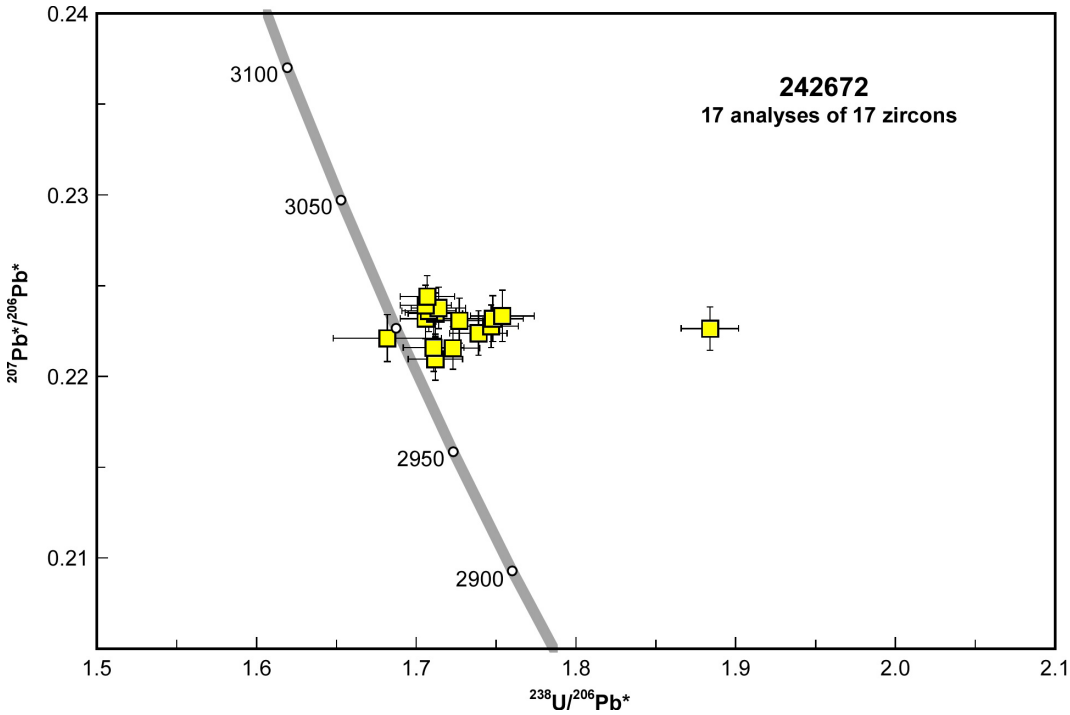

Figure 3. U–Pb analytical data for sample 242672: felsic metavolcanic rock, Solomon Road. Yellow squares indicate Group I (magmatic zircons)

Group I comprises 17 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3002 ± 4 Ma (MSWD = 0.62).

The date of 3002 ± 4 Ma for the 17 analyses in Group I is interpreted as the igneous crystallization age of the rhyolite.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 242672: felsic metavolcanic rock, Solomon Road; Geochronology Record 1800: Geological Survey of Western Australia, 4p.

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